Faculty of Engineering & Technology Electrical & Computer Engineering Department

Circuits and Electronics Laboratory - ENEE2103

Report #3

Exp 8 : The Field-Effect Transistor

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Section: 3
Date: 10/1/2023

Abstract:

This experiment introduces the field effect transistor. It intends to identify the field effect transistor characteristics and some applications on it such as amplifiers and constant current sources and some concepts were compared practically and virtually .

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1.Theory

1.1FET

The Field Effect Transistor (FET) is a three-terminal device. Three terminals are Drain Source (S) and Gate (G). In FET, current flow is due to only one type of charge (D), either electrons or holes. So, FET is known as unipolar device . with no PN- particles, junctions within the main current carrying path between the Drain and the Source terminals, the current path between these two terminals is called the channel which may be made of either a P-type or an N-type semiconductor material .

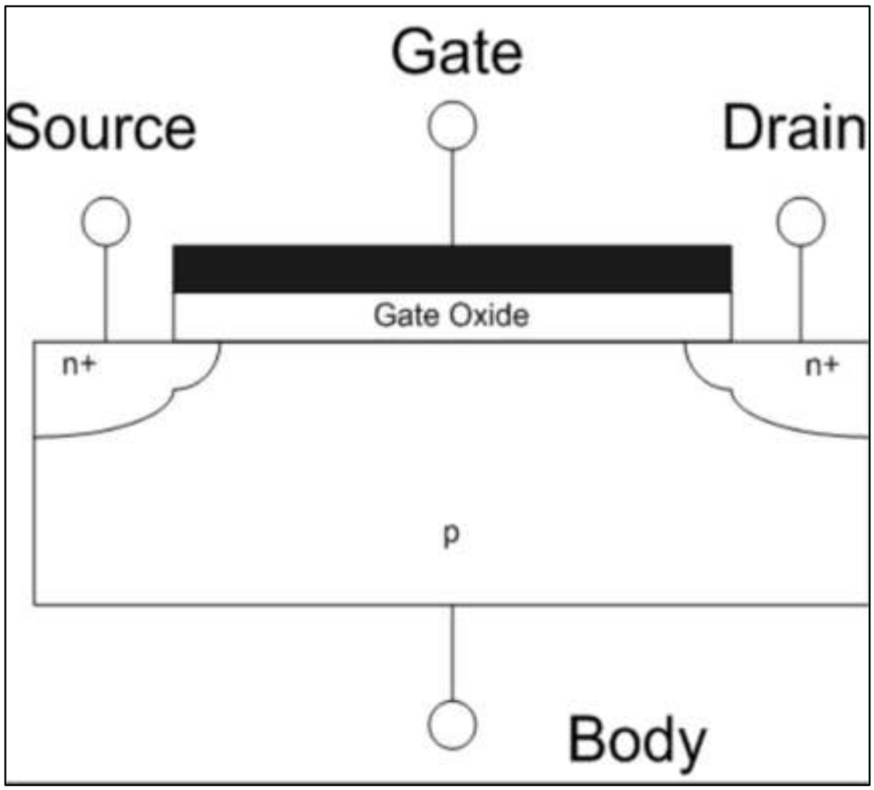


Figure 1.1.1 : FET terminals

There are two main types of field effect transistor, the Junction Field Effect Transistor (JFET) and the Insulated-gate Field Effect Transistor (IGFET), which is more commonly known as the standard Metal Oxide Semiconductor Field Effect Transistor or MOSFET for short .

Difference between BJT and FET

The main advantage of FET over bipolar junction transistor (BJT) is the input impedance in FET is very high compared to BJT. The other main advantage is BJT consumes more power and whereas FET utilizes less power for its operation, also need fewer step in manufacturing process.

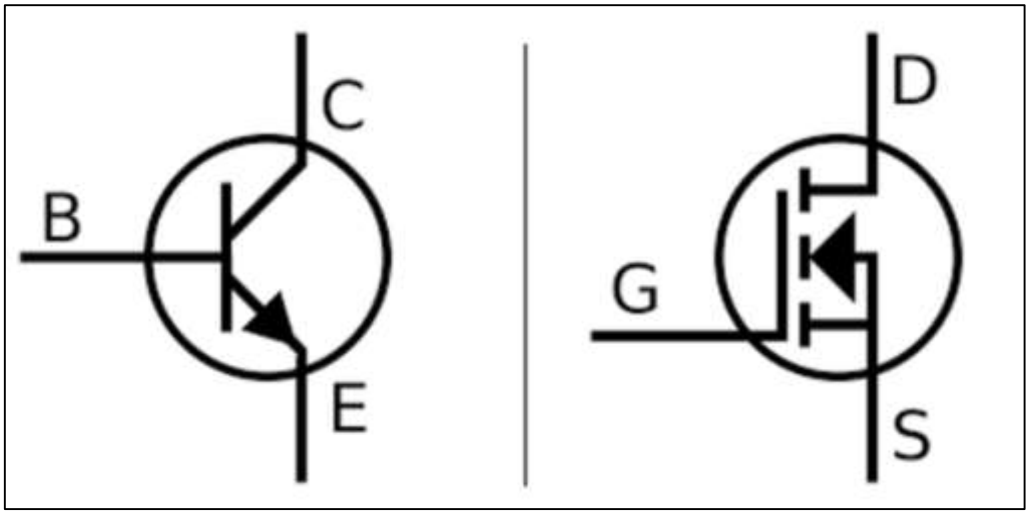


Figure 1.1.2 : BJT and FET Terminals

Working principle of FET

When a voltage V_{DS} is applied between drain and source terminals and voltage on the gate is zero, the electrons will flow from source to drain through a channel between the depletion layers. The size of the depletion layers determines the width of the channel and hence current conduction through the bar. However when a reverse voltage V_{GS} is applied between gate and source terminals the width of depletion layer is increased. This reduces the width of conducting channel so the resistance of n-type bar will increase (the current from source to drain is decreased. On the other hand, when the reverse bias on the gate is decreased, the width of the depletion layer also decreases. This increases the width of the conducting channel and hence source to drain current. (P-channel the same manner as an N- channel except that channel current carriers will be the holes instead of electrons and polarities of VGS and VDS are reversed).

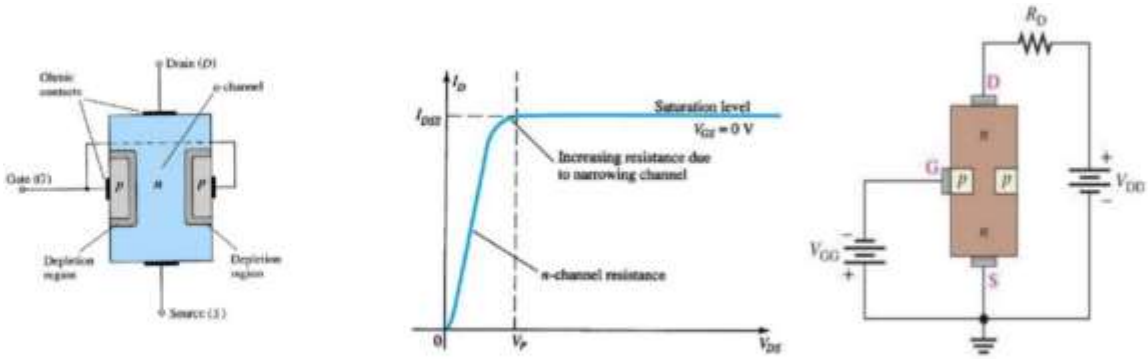


Figure 1.1.3 : JFET regions and characteristic

IDSS is the maximum drain current for a JFET and is defined by the conditions VGS = 0 V and VDS > |VP|.

we have four different regions of operation for JFET :

Ohmic Region happens when VGS = 0 the depletion layer of the channel is very small and the JFET acts like a voltage controlled resistor. Cut-off Region this region is also known as the pinch-off region were the Gate , voltage, VGS is sufficient to cause the JFET to act as an open circuit as the channel resistance is at maximum. Saturation or Active Region for these two regions the JFET becomes a good conductor and is controlled by VGS while VDS has little or no effect . and last Breakdown Region , in this region the voltage between the Drain and the Source is high enough to make JFET’s resistive channel breaking down and pass uncontrolled current .

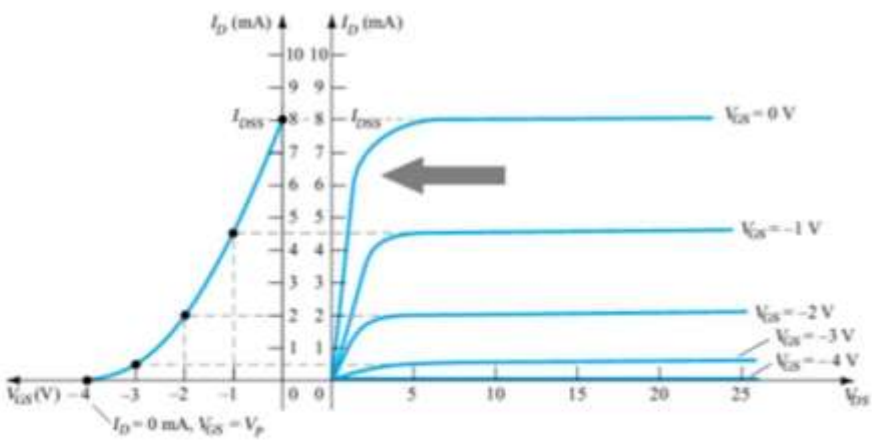


Figure 1.1.4 : JFET characteristic with all regions

2.Procedure

2.1 CHARACTERESTICS OF AN N-CHANNEL JFET.

- The circuit in the figure below was connected of taking into account the polarities.

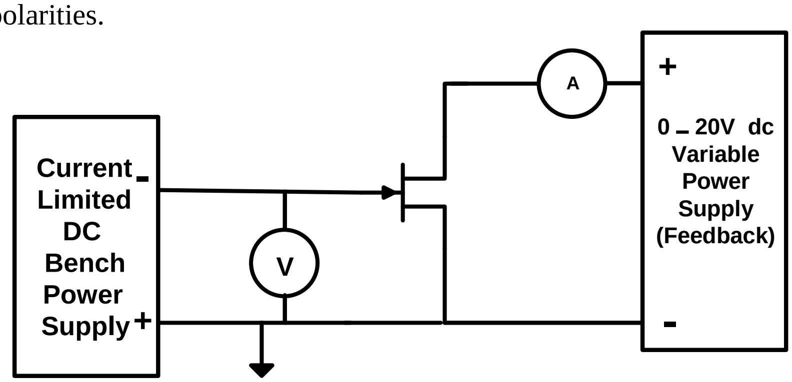


Figure 2.1 : Circuit Connection

- The current limit was set on the bench power supply to its minimum value .
- The voltage was set to zero , then switch on the power supply.
- For all the values of V_{DS}, the corresponding I_D values was recorded.
- V_{DS} was set to 10 V and V_{GS} to -1.0 V ,then I_G measured.

2.2 A JFET AMPLIFIER.

- the circuit was connected as shown in the figure below

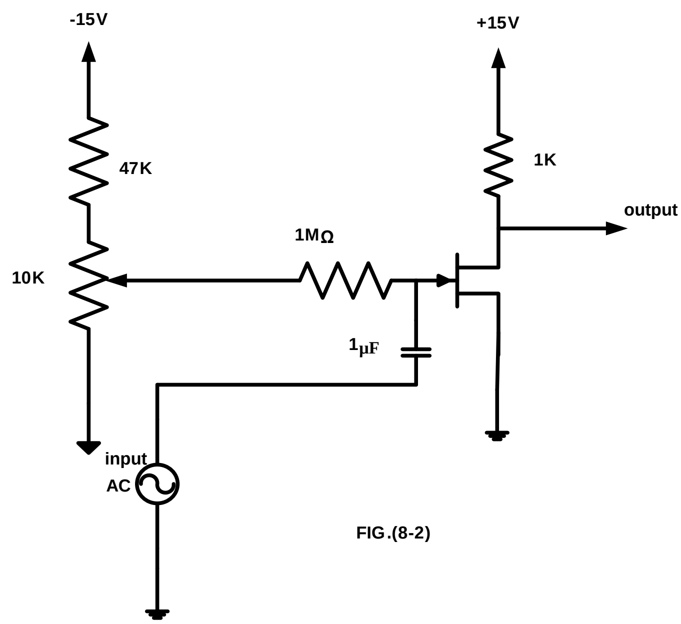


Figure 2.2 : Circuit Connection

- The sine wave generator was set to a frequency of 1 kHz ,but either disconnected its output .
- The potentiometer was set to give a value of +10 V for V_{DS}.
- An input was set of 2volts peak-to-peak from the generator and the output was observed on the oscilloscope.
- The peak-to-peak output voltage was measured and the voltage gain was calculated.
- The ac input current was measured and voltage using DMM and the input impedance Zin was calculated seen by the source.

2.3COMMON DRAIN AMPLIFIER

- The circuit was connected as shown the figure below.

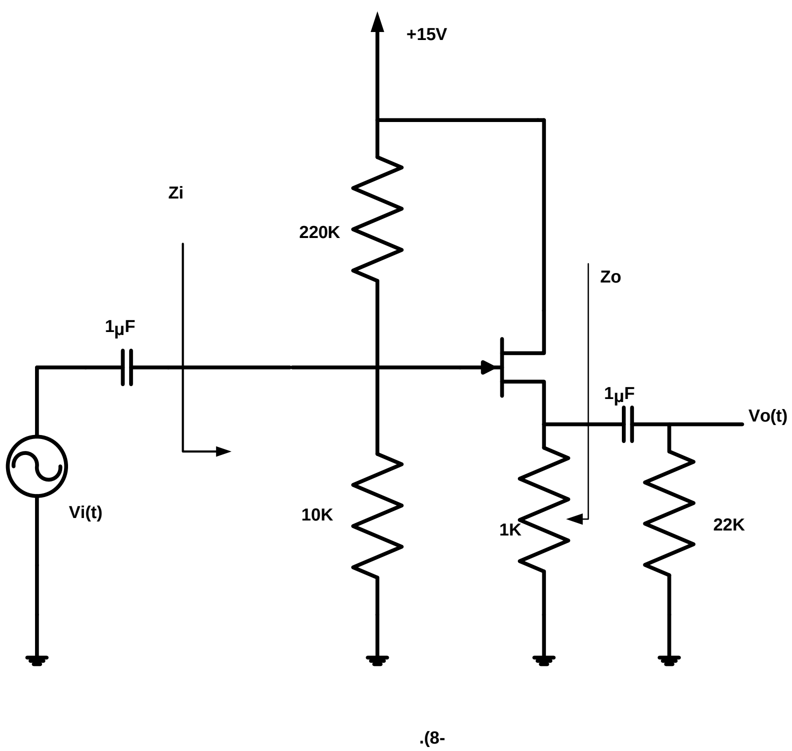


Figure 2.3 : Circuit Connection

- The sine wave generator was set to a frequency of 1 kHz ,but either disconnect its output .
- The DC voltages was measured of V_G, and V_S.
- An input was applied of 0.4 volts peak-to-peak from the generator and the output was observed on the oscilloscope.
- The voltage gain was calculated and the phase shift between the input and output voltage.
- The values of Z_{in} and Z_{out} was measured using the appropriate voltages and currents at the places shown in the previous figure.

2.4. CONSTANT CURRENT SOURCE

- The circuit was connected as shown the figure below.

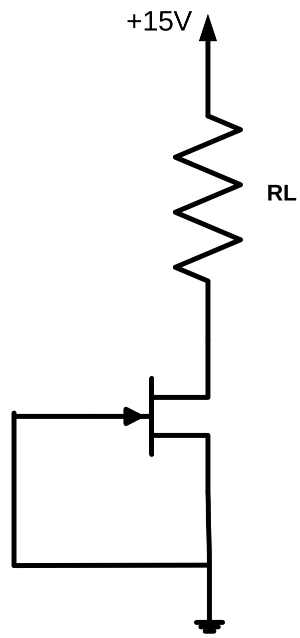


Figure 2.4 : Circuit Connection

- The values of the V_R was measured , and the current I_D calculated for each value of the resistor

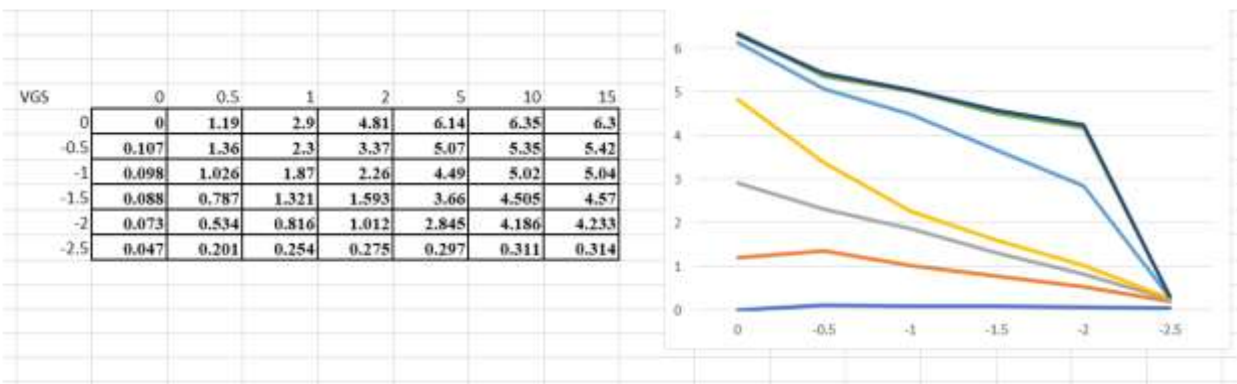
3. Data & Results

3.1 Characteristics of an N-channel JFET:

Table 3.1:The values of the characteristics of an N-channel JFET

V_{GS}(V)		I_D(mA) for V_{DS}=(V)						
Desired	**Actual**	**0**	**0.5**	**1**	**2**	**5**	**10**	**15**
0	**5uV**	**47uA**	**1.19**	**2.9**	**4.81**	**6.14**	**6.35**	**6.3**
-0.5	**-0.5**	**0.107**	**1.36**	**2.3**	**3.37**	**5.07**	**5.35**	**5.42**
-1.0	**-1**	**0.098**	**1.026**	**1.87**	**2.26**	**4.49**	**5.02**	**5.04**
-1.5	**-1.5**	**0.088**	**0.787**	**1.321**	**1.593**	**3.66**	**4.505**	**4.57**
-2.0	**-2**	**0.073**	**0.534**	**0.816**	**1.012**	**2.845**	**4.186**	**4.233**
-2.5	**-2.5**	**0.047**	**0.201**	**0.254**	**0.275**	**0.297**	**0.311**	**0.314**

Ig = 0.001mA



VGS	0	0.5	1	2	5	10	15
0	0	1.19	2.9	4.81	6.14	6.35	6.3
-0.5	0.107	1.36	2.3	3.37	5.07	5.35	5.42
-1	0.098	1.026	1.87	2.26	4.49	5.02	5.04
-1.5	0.088	0.787	1.321	1.593	3.66	4.505	4.57
-2	0.073	0.534	0.816	1.012	2.845	4.186	4.233
-2.5	0.047	0.201	0.254	0.275	0.297	0.311	0.314

Figure 3.1: Plot of Ids vs Vds

Discussion :

The Id stabilized the current at 10 volts of Vdc . Whenever Vgs = 0, Keep track of each time Vgs rises in magnitude minus the current Id falls whenever Vds reaches 10 volts. If Vds = 10 and Vgs = -1, you should be aware that Ig is far too low. When Vds increased by 0.5 in the negative, Id is reduced by 2.

Transconductance = gm=$\frac{\Delta ID}{\Delta VGS}$

3.2 JFET amplifier:

First, the circuit was connected as in figure 3.2.1 with resistors, JFET transistor, capacitor and power supply. Then we change the potentiometer to give VDS=10V. After that we set the generator to frequency 1 kHz and amplitude 2V and we get waves for input and output as shown in figure 3.2.2 .

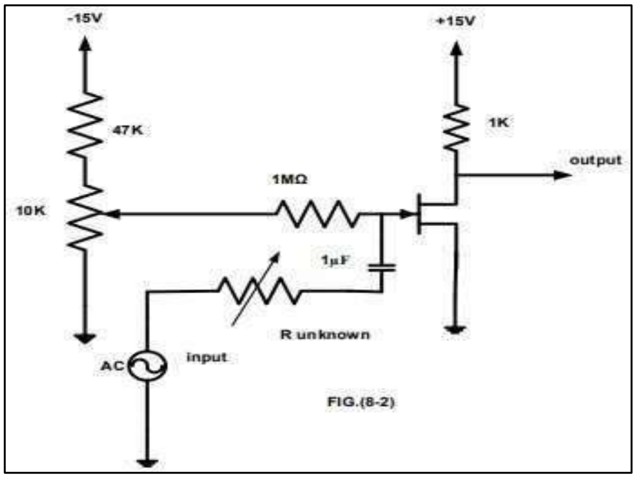


Figure 3.2.1 : Circuit connection

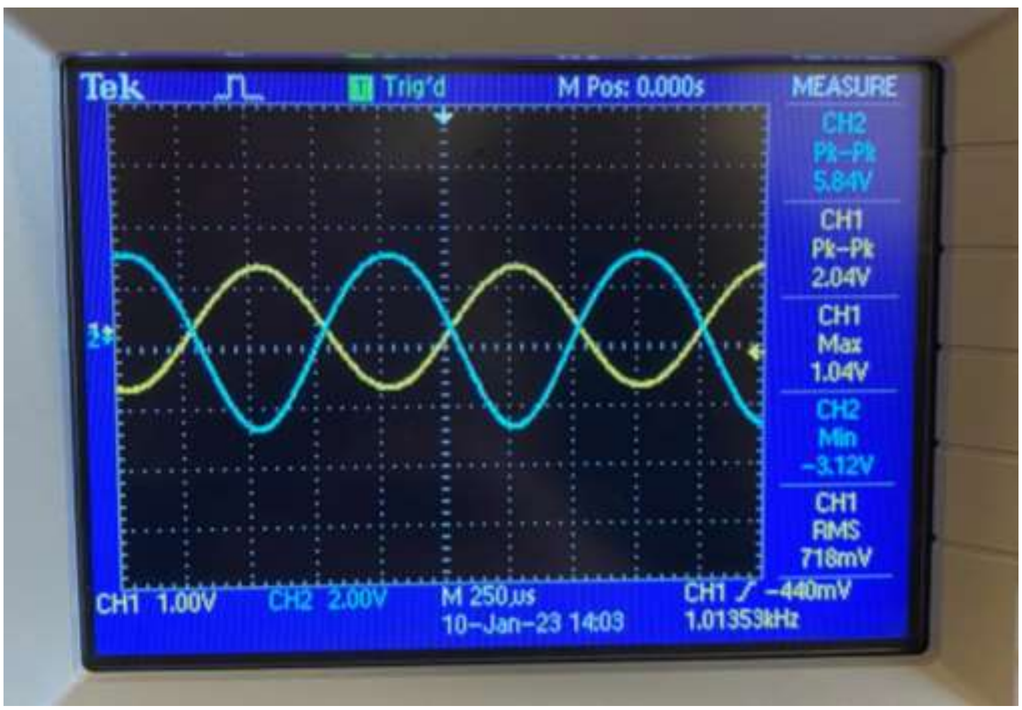


Figure 3.2.2 : input and output of circuit

Discussion :

Voltage gain = Vo / Vin = 5.84/2.04 = 2.86 .

Vi = 0.695 V , Ii = 5.7uA .

Input impedance = Vi / Ii = 0.695V/5.7uA = 121.9K ohm

3.3 Common drain amplifier :

the circuit was connected with resistors, JFET transistor and power supply. Then we set the generator to frequency 1 kHz and measure the DC voltage of VG and VS as we can see the VGS is reverse bias. After that we set the amplitude to 0.2V and we get waves for input and output as shown in figure 3.3 .

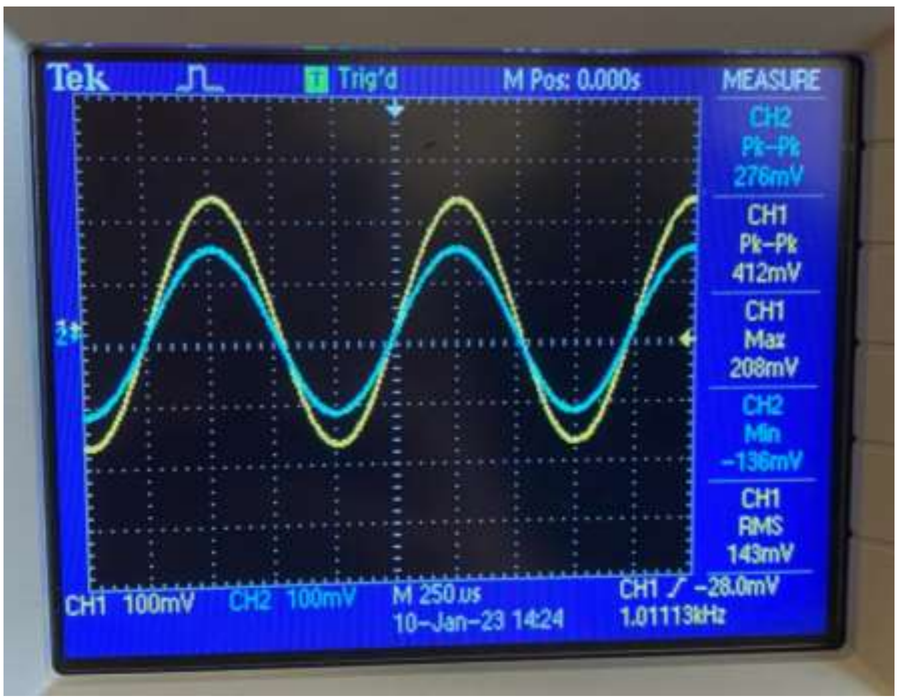


Figure 3.3 : input and output signals

Calculations :

The voltage gain = $\frac{V_{out}}{V_{in}} = \frac{280m}{408m}$ = 0.68V

The phase shift between the input and output voltage ≅ 0

Ii = 13.34µA & Io = 105.5 uV

Vi = 0.144v & Vo = 33.7mV

Zi = Vi/Ii = 0.144/13.34u = 108 KΩ

Zo = Vo/Io = 33.7m/105.5u = 319KΩ

Discussion:

As noted from the figure the peek to peek output is higher from the peek to peek input, and the phase shift between the input and output is almost zero. The theoretical and practical gain were very close and this indicates the success of the experiment. The theoretical one (0.7634) and the practical one (0.68).

3.4 : Constant current source:

the circuit was connected with variable resistors, JFET transistor and power supply. VRL and ID were measured by changing the resister between 0.1-3kΩ as shown in table 3.4 .

Table 3.4: Rl , Vl , ID table

R_L(KΩ)	V_L(V)	I_D(mA)
0.1	0.592	5.92
0.22	1.3	5.9
0.33	1.94	5.87
0.47	2.76	5.87
0.56	3.288	5.871
1	5.827	5.827
1.5	8.673	5.78
2	11.24	5.62
3	13.18	4.39

Discussion :

Shown in the above table, Id is almost constant (between 5.9 and 5.8). Which means that the JFET is in pinch off regions , and the voltage increased when the resistance was increased .

The results is almost close to practical values.

3 Conclusion :

This experiment was good in terms of introducing JFET and how it works using some of it’s applications , also it cleared up the difference between BJT and JFET and the use of each one . Moreover we understood the process of amplification and how FET is able to do it using it’s physical properties and how JFET works in different regions .

4 References

[1] lab manual accessed 12/1/2023 on 10:00 pm

[2] https://www.techtarget.com/whatis/definition/field-effect-transistor-FET

[3] https://www.electronics-notes.com/articles/electronic_components/fet-field-effect-transistor/what-is-a-fet-types-overview.php

Experiment #8 ENEE2103

The Field-Effect Transistor

Objectives:

1. To understand the difference between the bipolar and the field effect transistor.
2. To examine the characteristics of N-channel JFET when using as a common source and common drain.

Pre-lab Work:

1. Simulate the circuits in the procedure section and determine the required values (set the parameters that must be assigned by the instructor in the procedure to proper values).
2. Verify if Simulation Results match the expected results

Procedure:

I. CHARACTERESTICS OF AN N-CHANNEL JFET.

1. Connect the circuit of Fig.(8-1) taking into account the polarities.

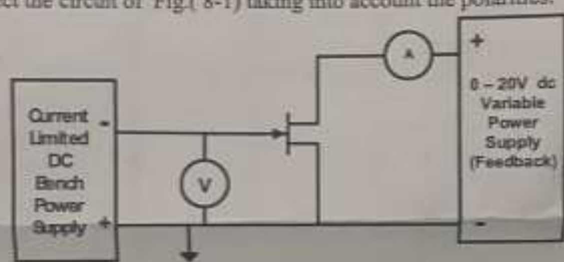


Fig. (8.1)

2. Set the current limit on the bench power supply to its minimum value .
3. Set the voltage to zero , then switch on the power supply.
4. Set the V_{DS} to the first value in table 8.1 , and then read I_D for each value of V_{GS}.
5. Repeat for all the values of V_{DS} in the table, recording the corresponding I_D values.
6. Plot the results from your table onto your graph, drawing one curve of I_D against V_{DS} for each value of V_{GS}.

Table 8.1

V_{GS}(V)		I_D(mA) for V_{DS}=(V)						
Desired	Actual	0	0.5	1	2	5	10	15
0	5m?	47 nA	1.14 mA	2.9	4.81	6.14	6.35	6.3
-0.5	0.5	0.107	1.36	2.3	3.37	5.07	5.35	5.42
-1.0	1	0.098	1.026	1.87	2.26	4.49	5.02	5.04
-1.5	1.5	0.088	0.787	1.321	1.593	3.66	4.505	4.57
-2.0	2	0.073	0.554	0.816	1.012	2.845	4.186	4.233

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-2.5	2.5	0.049	0.201	0.254	0.275	0.297	0.311	0.314

7. Now go back to your circuit and set V_{DS} to 10 V and V_{GS} to -1.0 V ,then try to measure I_G. $I_G = 0.001$

Note: When preparing the prelab in Pspice use dc and parametric sweep to get the curves measured in Table 8.1

Questions:

- From your graph above which values of V_{DS} is I_D almost unaffected by V_{DS} when V_{GS}=0?
- For a given value of V_{DS} , (say 10 V),do equal changes of V_{GS} cause equal changes of I_D?
- Can you measure I_G or is it too small?
- From your graph, estimate the change in I_D for 0.5 change in V_{GS} when V_{DS} =10 V , and V_{GS} -1.0 V ,then find the trans conductance of the transistor(g_m).

Note: trans-conductance g_m =(change in I_D)/(change in V_{GS}).

II. A JFET AMPLIFIER.

1. Connect the circuit as shown in Fig. (8-2).

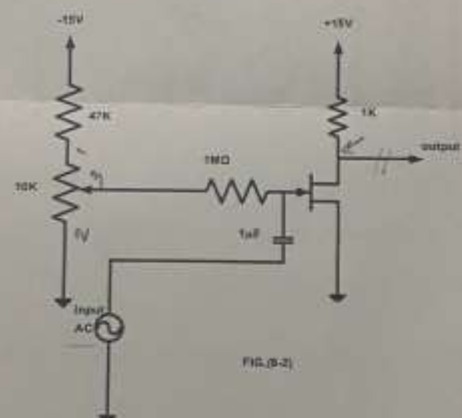

FIG.(8-2)

2. Set the sine wave generator to a frequency of 1 kHz ,but either disconnect its output ,or turn its output amplitude to zero, so there is no signal input to the circuit.

3. Set the potentiometer to give a value of +10 V for V_{DS}.

4. Now apply an input of 2volts peak-to-peak from the generator and observe the output on the oscilloscope.

5. Measure the peak-to-peak output voltage and calculate the voltage gain.

5.7μA 0.695 volt

6. Measure the ac input current and voltage using DMM and calculate the input impedance Zin seen by the source.

III. COMMON DRAIN AMPLIFIER.

1. Connect the circuit as shown in fig. (8-3).

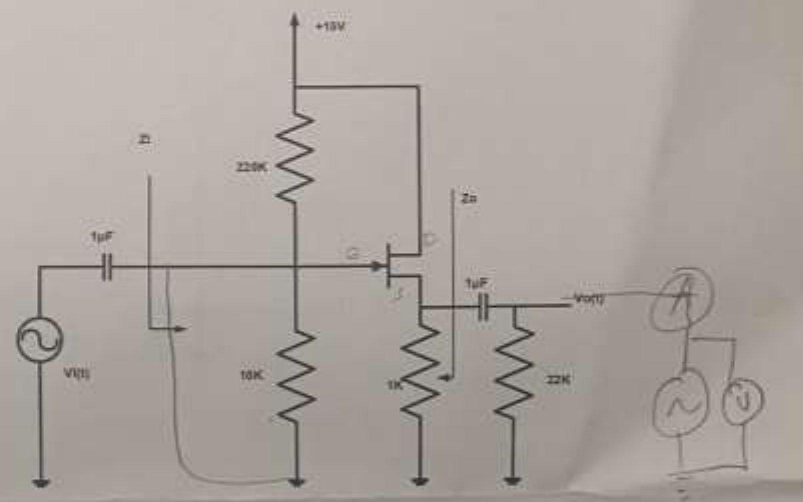

Fig.(8-3)

2. Set the sine wave generator to a frequency of 1 kHz ,but either disconnect its output ,or turn its output amplitude to zero, so there is no signal input to the circuit.

3. Measure the DC voltages of V_O, and V_S. $V_e = 0.643v$ $V_s = 2.08v$
4. Now apply an input of 0.4 volts peak-to-peak from the generator and observe the output on the oscilloscope. 408mV $V_o = 250mV$
5. Calculate the voltage gain and the phase shift between the input and output voltage. 0 inphase

6. Measure the values of Z_{in} and Z_{out} using the appropriate voltages and currents at the places shown in the previous figure. $V_{in} = 0.141$ $I_i = 13.34\mu A$ | $V_o = 33.7mV$ $I_o = 85.5\mu A$

Question:

- Compare the voltage gain of step 5 with the theoretical gain value.

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III. CONSTANT CURRENT SOURCE.

- Connect the circuit as shown in fig.(8-4).

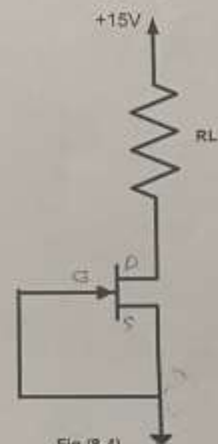

Fig.(8-4)

- Measure the values of the V_R ,and calculate the current I_D for each value of the resistor, then record your result in a table like table 8.2

Table 8.2

R_L(KΩ)	V_L(V)	I_D(mA)
0.1	0.592	5.92
0.22	1.3	5.9
0.33	1.94	5.87
0.47	2.76	5.87
0.56	3.288	5.871
1	5.827	5.827
1.5	8.673	5.78
2	11.24	5.62
3	13.18	4.39

$I_D = \frac{V_L}{R_L}$

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Question:

- Calculate the maximum theoretical value of load resistance for which the JFET- current source can provide constant current.
- Comment on the measured range of load resistance in comparison to the theoretical range of load resistance.